

Complete sequences are weighted lower semi-frames

Literature-implied status packet for arXiv:2606.00764

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Open question

The source paper is E. Zikkos, *The Gaussian Gabor system at the critical density is a weighted lower semi frame*, arXiv:2606.00764. It quotes Conjecture 3.1 of P. Balazs, M. Corso, and D. T. Stoeva, *Weighted frames, weighted lower semi frames and unconditionally convergent multipliers*, arXiv:2310.18957.

The conjecture asks whether every complete family in a separable Hilbert space is a weighted lower semi-frame. The source paper proves the case where the family contains a complete minimal subfamily, then leaves open the case in which every complete subfamily has infinite excess. It also singles out the Muntz family $\{t^{\lambda_n}\} \subset L^2(0, 1)$, with $\sum_n \lambda_n^{-1} = \infty$, as an unknown example.

Theorem 1.2 answers partially the following conjecture.

Conjecture 1.1. *[1, Conjecture 3.1] If a family $\mathcal{F}$ in a separable Hilbert space $\mathcal{H}$ is complete, then $\mathcal{F}$ is a weighted lower semi frame for $\mathcal{H}$.*

The conjecture is known to hold if $\mathcal{F}$ has a reproducing partner for $\mathcal{H}$ ([1, Proposition 3.3]). A special case is when $\mathcal{F}$ contains a Schauder basis for $\mathcal{H}$ or a frame for $\mathcal{H}$ ([1, Corollary 3.5]). Corollary 1.2 adds a further case: any complete family containing a complete and minimal subfamily.

Remark 1.4. *The case when $\mathcal{F}$ is complete but every complete subfamily has infinite excess remains open.*

A natural example is the Müntz family $\{t^{\lambda_n}\}$, where $\{\lambda_n\} \subset \mathbb{N}$ and $\sum \lambda_n^{-1} = \infty$. By the Müntz-Szász theorem the family is complete in $L^2(0, 1)$ if and only if $\sum \lambda_n^{-1} = \infty$. Thus, every complete subfamily of $\{t^{\lambda_n}\}$ again has an infinite excess in $L^2(0, 1)$. Whether such families are weighted lower semi frames for $L^2(0, 1)$ is unknown.

Literature-implied answer

N. Ozawa's preprint *Total sequences and the lower frame inequality*, arXiv:2507.04899, proves exactly the required Hilbert-space statement. Ozawa states the theorem as a solution to a Math-Overflow exercise, not as a solution of the Balazs–Corso–Stoeva or Zikkos formulation; the match below is the reformulation recorded by this packet.

Theorem (Ozawa, arXiv:2507.04899). *Let $(v_n)_{n=1}^\infty$ be a total sequence in a Hilbert space H . Then there is a sequence $(\lambda_n)_{n=1}^\infty$ of positive numbers such that*

$$\|x\|^2 \leq \sum_{n=1}^{\infty} \lambda_n |\langle x, v_n \rangle|^2$$

for every $x \in H$, with the right-hand side possibly equal to ∞ .

This is precisely the lower semi-frame inequality after scaling: define

$$\omega_n = \sqrt{\lambda_n}.$$

Then

$$\sum_{n=1}^{\infty} |\langle x, \omega_n v_n \rangle|^2 = \sum_{n=1}^{\infty} \lambda_n |\langle x, v_n \rangle|^2 \geq \|x\|^2$$

for every $x \in H$. Thus $(\omega_n v_n)$ is a lower semi-frame with lower bound 1. If the BCS formulation requests an arbitrary prescribed lower bound $A > 0$, replace λ_n by $A\lambda_n$.

Conclusion. *Ozawa's theorem fully answers the conjecture quoted in arXiv:2606.00764 affirmatively. In particular, the Muntz-family example in the source paper is weighted lower semi-frame whenever it is complete.*

Proof idea from Ozawa

Let

$$H_n = \overline{\text{span}}\{v_k : k \geq n\}, \quad H_\infty = \bigcap_n H_n.$$

The descending chain $H_1 \supset H_2 \supset \dots$ has one-dimensional successive quotients at most:

$$\dim(H_n \cap H_{n+1}^\perp) \leq 1.$$

Choose an orthonormal basis from these quotient directions together with an orthonormal basis of H_∞ , ordered so that the n -th basis vector lies far enough in the tail space $H_{\lceil n/2 \rceil}$. Approximate those basis vectors by finite linear combinations of the tail vectors v_k . The approximants define an operator that is a small-norm perturbation of the identity, hence invertible. Expanding the resulting strong-operator identity back into the original v_k 's and applying Cauchy–Schwarz produces the coefficients λ_k .

Status

This is a literature-implied answer packet, not an original pipeline solution. The earlier partial packet proving the countable complete-block criterion remains valid, but it is superseded for the full conjecture by arXiv:2507.04899.